GAÉTAN CANTELOBRE

EMBEDDED SYSTEMS ENGINEER

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github.com/gaetancantelobre | linkedin.com/in/gaetan-cantelobre | Dual Nationality: French / American (Eligible for Clearances)
Master's in Embedded Software Development — Université de Bordeaux

Embedded Systems Engineer specializing in C/C++, FreeRTOS, and custom electronics design for Avionics, Defense, and Autonomous Systems. Experienced in aerospace and defense industry standards (Thales, Delair) across the full product lifecycle, from schematic capture and board bring-up to deterministic flight firmware and automated qualification benches.

PROFESSIONAL EXPERIENCE

Embedded / Autopilot Engineer — DELAIR, Toulouse

April 2026 – Present | CDI

Autopilot Software Development & GNC

- Developed autonomous flight control modes in FreeRTOS, including automated waypoint navigation, maritime vessel landings, and evasive maneuvers.
- Enforced software quality, reliability, and cybersecurity compliance in strict adherence to ISO/IEC 25010 and ISO/IEC 27001 standards.
- Engineered flight control and mission systems for both loitering munitions and long-range intelligence, surveillance, and reconnaissance (ISR) UAVs.
- Implemented Extended Kalman Filter (EKF) state estimation for robust navigation under nominal and GPS-denied environments.

Advanced Software Testing & Flight Analysis

- Conducted extensive validation campaigns across Software-in-the-Loop (SIL), Hardware-in-the-Loop (HIL), and live in-flight trials.
- Integrated custom machine learning and telemetry services for automated flight data recorder (black box) pre-analysis, accelerating diagnostic workflows.

Technologies: C++, FreeRTOS, EKF, SIL/HIL Testing, Python, Linux, UAV Avionics, Telemetry Analysis.

Embedded / Real-Time Developer — THALES SIX GTS, Cholet

April 2025 – September 2025 | Master's Internship

Timing Driver Development & System Integration

- Architected and implemented a microsecond-precision timing driver for military tactical radio test systems using C++ (11/17) and Python, purpose-built for HIL testing benches.
- Optimized deterministic execution by integrating the timing driver into a real-time OS environment, reducing test cycle latency and maximizing throughput.

Debugging & Environment Optimization

- Modernized legacy debugging procedures by reintegrating specialized diagnostic tools, delivering measurable reductions in troubleshooting time for complex embedded faults.
- Enhanced diagnostic reliability by implementing automated telemetry logging and fault-tracking scripts in Python, increasing system observability.

Technologies: C++11/17, Python, In-house RTOS, Linux, Embedded Systems, HIL Qualification.

TECHNICAL SKILLS

Languages:	C, C++ (11/17), Python, Bash, Inline Assembly
RTOS & Kernels:	FreeRTOS, In-house RTOS, Embedded Linux, Bare-metal
Platforms & MCUs:	STM32 (F4/G4), ESP32, RP2040, ATmega, NVIDIA Jetson Nano
Hardware & PCB:	KiCad (Schematic & Layout), SMD Soldering & Hot-Plate Reflow, BMS Design
Tools & Methods:	HIL/SIL Testing, GDB, Oscilloscopes, Logic Analyzers, Git, Docker, Make/CMake

KEY ENGINEERING PROJECTS

Drone Défense Hackathon (Challenge 4) — French Army Logistics

Paris (2025)

- Architected an autonomous drone delivery and tactical logistics infrastructure concept for forward military operations.
- Implemented A* and RRT algorithms for real-time trajectory optimization and collision avoidance.
- Developed responsive command interfaces and live tracking dashboards for real-time mission monitoring and control.

Machine Learning-Enhanced Digital Level — Competitive Shooting

Hardware & Firmware

- Engineered custom ESP32-C3 PCB featuring an LSM6DSL 6-axis IMU, magnetic WS2812B LED array, and custom Li-ion BMS.
- Implemented on-device TinyML classification to detect rifle cant instability and erratic barrel movement prior to trigger break.
- Developed real-time firmware in C++ using ESP-IDF, maximizing battery life through deep sleep power management.

Custom UAV Design & Flight Controller Development — Avionics

Hardware & Firmware

- Designed aerodynamic 3D-printed aircraft structures in OpenVSP and Autodesk Fusion 360, achieving sub-250g AUW limits.
- Engineered 3 custom STM32G4-based flight controller boards running custom FreeRTOS firmware for fixed-wing and multirotor platforms.
- Integrated ExpressLRS (ELRS) serial communication protocols for low-latency command link and telemetry transmission.